JCB EXCHANGE

# Platform Change Report

A client-ready overview of the latest backend, website and operations portal improvements.

REPORTING DATE 03 November 2026
STATUS Pre-release review
SCOPE Backend, public website, admin, employee and partner portals

| | |
|---|---|
| Prepared for | **JCB Exchange Client Team** |
| Purpose | **Change summary and release readiness** |

EXECUTIVE OVERVIEW

# A broad platform improvement release

This work is a substantial upgrade focused on safer operations, better vehicle-listing controls, clearer verification history, and stronger sales and CRM workflows.

| 181 | 47 | 3 | 0 |
|---|---|---|---|
| Approx. affected paths | New files | Production builds passing | Mobile app code changes |

**Overall position: strong feature progress, final stabilization pending**

The main systems compile successfully and most automated checks pass. A small number of release blockers must still be resolved before deployment, including the partner Add/Edit Vehicle modal and admin quality-gate failures.

## What this release delivers

### Safer destructive actions

Users can see what will be deleted, retained, unlinked or blocked before a permanent deletion. Financial and audit records receive stronger protection.

### Vehicle identity control

Registration and chassis numbers are normalized and checked to reduce duplicate listings, including protection against simultaneous submissions.

### Clear listing lifecycle

Listings can be unlisted, suspended, blocked, restored and sent for review with reasons, permissions and audit history.

### Unified verification history

Identity and vehicle checks can be reviewed through consolidated, searchable and permission-aware history screens.

## Change footprint

| Area | Earlier challenge | Current solution |
|---|---|---|
| **Backend** | Deletion, listing identity, moderation and history rules required stronger central control. | New domain services, routes, migrations, rate limits and audit-aware workflows. |
| **Public website** | Seller forms could lose progress and discover duplicate vehicles too late. | Draft restore, live duplicate hints, moderation actions and safer listing deletion. |
| **Operations portal** | CRM, RTO, verification and deletion workflows were fragmented. | Major workflow redesigns, shared controls, history pages and readable routes. |
| **Mobile app** | No mobile change was included in this work package. | No source impact from this change set. |

BUSINESS IMPACT

# Problems addressed and solutions delivered

The following changes are grouped by the business risk or operational gap they resolve.

| Area | Earlier challenge | Current solution |
|---|---|---|
| **Permanent deletion** | Operators could remove records without a complete view of linked data. | Impact preview, required confirmations, optional file cleanup and safer alternatives. |
| **Financial history** | Sales and RTO deletions could conflict with receipts, bills, refunds or posted finance entries. | Money-linked records block deletion; Cancel Sale preserves required history. |
| **Duplicate vehicles** | Formatting differences and race conditions could allow the same vehicle to be listed twice. | Normalized keys, live checks, transactional recheck and indexed lookup. |
| **Listing moderation** | Hiding or restricting a listing lacked a controlled lifecycle and clear seller recovery. | Unlist, Suspend and Block states with reasons, audit records and review actions. |
| **Partner CRM access** | Incomplete partner registration could incorrectly grant CRM access. | CRM access now requires an approved partner profile. |
| **Contact privacy** | Partner lookup could expose unrelated contacts. | Mobile lookup is limited to contacts already shared through the partner's leads. |
| **Verification records** | History was split across multiple identity and vehicle services. | Unified search, filters, totals and paginated history by permission. |
| **Public buyer privacy** | Sold-listing data could expose more buyer information than necessary. | Buyer name is masked on the server before it reaches the browser. |

### Important control principle

The backend rechecks deletion impact, permissions and duplicate vehicle identity at the moment of saving or deleting. The system does not rely only on what the user interface displayed earlier.

LISTINGS AND SALES

# Vehicle listing controls and seller experience

The listing workflow now combines better fraud prevention, seller guidance and operational visibility.

### Duplicate vehicle prevention

Registration and chassis values are standardized, indexed and checked on both live entry and final save. Admin users can review duplicate and resale context.

### Moderation lifecycle

New Unlisted, Suspended and Blocked states support owner/admin actions, reasons, notifications and audit records.

### Seller draft recovery

The public Sell Vehicle form can preserve and restore an in-progress draft, reducing accidental data loss.

### Safe listing deletion

Enquiries and RTO history are retained and unlinked. Sales, payments and bills prevent destructive deletion. Unlist is offered as an alternative.

### Blocked-vehicle protection

A blocked listing cannot be edited by the seller, linked to a sale or relisted under the same vehicle identity until an admin unblocks it.

### Canonical public links

Dealer and machine links now use readable, consistent paths, including return navigation after payment.

## Sold Vehicle and RTO integrity

- Sold Vehicle and connected RTO records now use a shared impact-aware deletion workflow.
- Issued receipts, bills, refunds, seller payments, locked records and posted finance entries protect the sale record from deletion.
- Cancel Sale preserves financial history while undoing operational side effects safely.
- A listing returned from a cancelled or deleted sale remains hidden until it is reviewed and deliberately relisted.
- RTO screens now show expense, advance and balance information and provide contextual customer, seller and agent communication shortcuts.

CRM AND CUSTOMER OPERATIONS

# A more complete operational workspace

The admin and employee experience has been expanded from isolated records into clearer customer, campaign and performance views.

### Won Customers

A dedicated directory shows customers with at least one Won CRM deal, with unique rows, search, pagination, Prime status, owner, location and latest enquiry information.

### Customer 360 view

The previous visitor detail page is replaced by the richer CRM contact detail experience, reducing duplicate interfaces and giving teams one consistent customer view.

### CRM dashboard

Agenda, inquiry, pipeline and source metrics are presented in a denser operational dashboard with search and filters.

### Campaign management

Smart audience presets, audience estimation, table/card views, sample inspection and WhatsApp dispatch preview improve campaign preparation.

### Reports and conversion

Conversion tables, stage/source analysis, trend views and calculated KPIs provide a clearer view of pipeline performance.

### Listing analytics

Traffic, enquiries, commercial performance, funnel health, dealer origin and data-integrity signals are consolidated into one listing detail report.

## Access and privacy improvements

- Only approved partner accounts can open partner CRM.
- Partner contact lookup is limited to the partner's own shared lead network.
- Readable contact and lead links are supported without exposing full raw identifiers in normal navigation.
- Customer pages are available in both admin and permission-based employee portals.

VERIFICATION AND DOCUMENTS

# Identity and vehicle checks brought together

The latest change set gives operations teams a unified way to review verification activity while keeping service-level permissions intact.

| Area | Earlier challenge | Current solution |
| --- | --- | --- |
| **Identity Hub** | PAN, GST, Aadhaar and Mobile-to-PAN activity was split across separate views. | Unified overview, detail workflows, history filters and saved result/template access. |
| **Vehicle Hub** | Driving licence, RC, challan, chassis and mobile checks lacked one consolidated history. | Unified vehicle history, recent activity and dedicated full-history pages. |
| **Employee access** | History visibility needed to follow each verification permission. | Employees see only permitted verification types; admins can see the full set. |
| **Document previews** | PAN, Aadhaar and RC preview tools were exposed through public-site routes. | Templates moved to authenticated admin document-template routes; duplicate public assets were removed. |
| **Usage protection** | Paid KYC and RC lookup operations needed stronger cost and abuse controls. | Account-level and daily rate limits were added, with appropriate staff handling. |

## History capabilities

- Newest-first combined timeline across supported verification services.
- Server-side search, service filters, pagination and per-service totals.
- Stale-response protection when users switch filters or pages quickly.
- Dedicated admin and employee routes plus recent-history panels inside each hub.
- The combined all-service view is capped at 2,000 reachable records; selecting one service supports deeper source pagination.

PLATFORM FOUNDATION

# Reliability, security and operational add-ons

Several supporting improvements make the new business workflows safer to operate and easier to maintain.

### API error handling

Duplicate listings, blocked deletion, database link conflicts and Drive failures now return clearer response codes and messages.

### Rate limiting

Vehicle identity, paid partner KYC and seller/dealer RC lookup endpoints receive account-aware limits.

### Recruitment history

Jobs with applications cannot be permanently deleted; teams are directed to close or archive them.

### Role synchronization

A new operations script standardizes Manager, CRE, Accountant, CA, HR and SEO Manager roles and cleans legacy naming.

### Fresh database setup

A guarded script can initialize an empty database where a traditional migration baseline is not available.

### Readable navigation

Customer, finance, listing payment, Sold Vehicle and RTO records use clearer routes with safe short-ID resolution.

### Shared confirmations

Portal delete actions use a consistent confirmation experience instead of inconsistent browser prompts.

### Frontend connectivity

A development API fallback is included in the content security policy when the public API environment variable is absent.

### Database deployment note

The new listing identity and moderation migrations must be included and applied before deploying application code that reads the new fields and statuses.

RELEASE READINESS

# Current validation and remaining work

The feature set is advanced, but this document intentionally separates completed implementation from release readiness.

| Product | Build | Automated checks | Release position |
|---|---|---|---|
| **Backend** | Pass | 637 of 641 tests pass | Needs clean test output |
| **Public frontend** | Pass | 43 of 43 tests pass | Green in current automated checks |
| **Admin portal** | Pass | 250 of 255 tests pass; lint has 28 errors | Requires fixes before release |
| **Mobile app** | No code change | Outside this change package | No release impact from this patch |

### Critical blocker - Partner Add/Edit Vehicle

The old partner listing modal is disabled, while its new replacement is currently an empty placeholder. In the present code state, the partner Add/Sell/Edit Vehicle form does not open. This must be completed or the previous modal must be restored before release.

## Other items to close before deployment

- Resolve 28 admin lint errors and review the remaining warnings in changed files.
- Resolve five admin test failures covering routing, templates, service-card behavior and generated KYC documents.
- Clean the backend build output before tests so four obsolete compiled tests are not executed.
- Add all currently untracked migrations, backend services and UI components to the intended commit set.
- Remove the unused admin configuration import and normalize line endings to reduce future diff noise.
- Run manual smoke tests for listing create/edit/delete/moderation, account deletion, Sold Vehicle/RTO cancellation and verification history.

RECOMMENDED PLAN

# A controlled path to release

The following sequence keeps database, backend and interface dependencies aligned.

| | |
|---|---|
| 01 | **Restore the partner vehicle form**<br>Complete the extracted Add/Edit Vehicle modal or temporarily restore the proven implementation. |
| 02 | **Validate database migrations**<br>Apply both migrations to a disposable database and review the duplicate-vehicle backfill in dry-run mode. |
| 03 | **Finalize backend services**<br>Commit the deletion, identity, moderation and verification-history services together with routes and tests. |
| 04 | **Finalize portal and website integration**<br>Include all new dialogs, pages, hooks, route helpers and template moves as one complete dependency set. |
| 05 | **Close quality gates**<br>Fix admin lint and tests, clean backend output and rerun all automated checks from a clean state. |
| 06 | **Run business smoke tests**<br>Validate high-risk workflows with realistic accounts, listings, sales, payments and verification data. |

**Client takeaway**

This release meaningfully improves operational safety, vehicle-data integrity, customer management and reporting. Once the listed stabilization items are closed, it will represent a strong foundation for the next production deployment.

## Document note

This report summarizes the current uncommitted change set as reviewed on 03 November 2026. It is a pre-release communication document, not a production completion certificate.